



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SSR FUEL SERVICE INDIAN OIL D

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

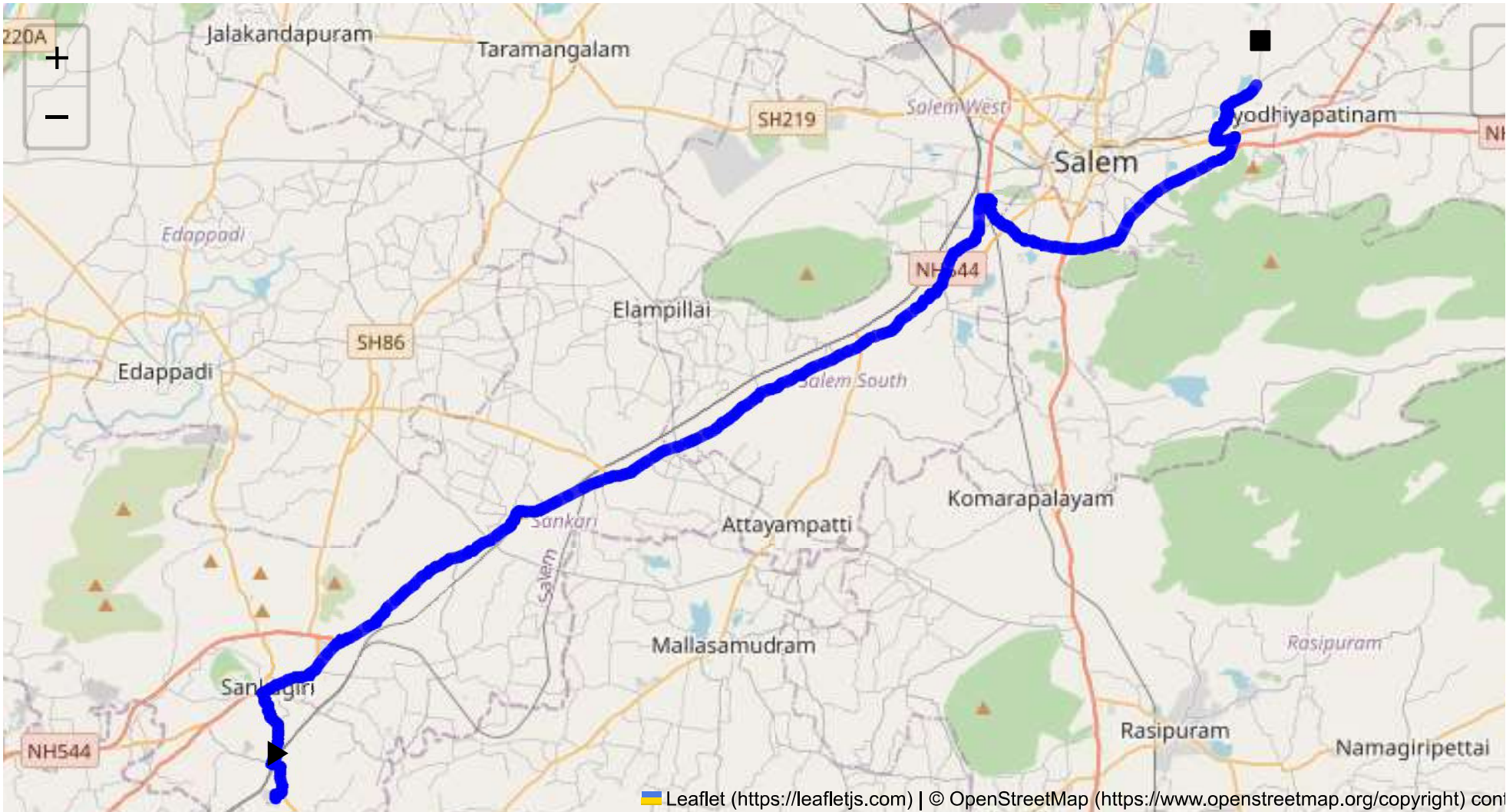
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 55.09 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (11.680881, 78.214074)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Veeranam, Tamil Nadu

1. Overview of the Route Map

The route spans approximately 55.09 kilometers, starting from Sangagiri and ending in Veeranam. It primarily follows major roads, likely involving State Highways and National Highways, which are typical

for inter-region travel in Tamils Nadu.

2. Typical Weather Conditions and Potential Hazards

Tamil Nadu generally experiences tropical weather with high temperatures. The region can encounter heavy monsoons from October to December, potentially causing floods or waterlogged roads. Heatwaves are common in summer months, which may affect vehicle cooling systems and tire conditions.

3. Traffic Patterns

Traffic tends to be heavier during morning (7:30 AM - 10:00 AM) and evening (4:00 PM - 7:00 PM) peak hours. Key congestion-prone areas include entry and exit points to major towns and intersections leading to commercial zones or market areas.

4. Road Quality and Infrastructure

The major highways are typically well-paved and maintained; however, local roads leading to villages might have potholes and narrower lanes. Road signage may be missing in rural stretches, and there may be unmarked speed bumps.

5. Alternative Routes for Emergencies

In case of roadblocks, taking a detour via the next closest National or State Highway is advisable. Ensure to have updated maps as local roads may vary in quality and accessibility.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials in Tamil Nadu requires compliance with the Motor Vehicles Act, ensuring vehicle and driver licensing meet stipulations for dangerous goods. Certain materials may have restricted hours for transport through densely populated areas.

7. Historical Incidents Involving Heavy Vehicles

There have been incidents of overturns due to sharp turns and accidents at congested intersections. Most are attributed to speeding or failure to adhere to local traffic regulations.

8. Environmental Considerations and Sensitive Areas

The route may pass near agricultural lands and rivers, particularly sensitive to chemical spills. Maintaining containment measures is crucial to prevent environmental contamination.

9. Communication Coverage and Potential Dead Zones

Most highways in Tamil Nadu maintain good mobile network coverage. Rural connections may have occasional dead zones, primarily in hilly or forested areas.

10. Estimated Emergency Response Times

Emergency services might take 30 to 60 minutes, depending on proximity to major towns. Sangagiri and Salem will generally offer quicker response times due to the presence of more resources.

11. (Corrected: This was mistakenly labeled as point 12 in your input)
Overall Summary of Risk Assessment

With regular traffic flow interrupted by intermittent congestion and under certain weather conditions, risks involve road quality and historical precedents of mishaps involving heavy vehicles. Measures include timing travel to avoid peak hours, ensuring vehicles are in prime condition, and drivers are informed of potential hazards. Emergency plans and alternative routes should likewise be strategically pre-planned.

In conclusion, drivers should keep vigilant, especially during changing weather conditions and while navigating transitions between rural and urban infrastructure. Constant communication and adherence to safety protocols will mitigate most of the potential risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45004, 77.87206	15 KM/Hr
5	Turn	Medium	11.45028, 77.87202	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	High	11.49149, 77.89601	15 KM/Hr
8	Turn	High	11.49224, 77.89601	15 KM/Hr
9	Turn	Medium	11.61736, 78.10588	30 KM/Hr
10	Turn	Medium	11.61753, 78.10608	30 KM/Hr
11	Turn	Medium	11.64162, 78.12160	30 KM/Hr
12	Turn	Medium	11.64120, 78.12168	30 KM/Hr
13	Turn	Medium	11.63948, 78.12081	30 KM/Hr
14	Turn	Medium	11.65627, 78.20215	30 KM/Hr
15	Blind Spot	Blind Spot	11.66372, 78.20649	10 KM/Hr
16	Blind Spot	Blind Spot	11.66216, 78.19911	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Turn	High	11.66365, 78.19907	15 KM/Hr
18	Turn	Medium	11.66615, 78.20042	30 KM/Hr
19	Turn	Medium	11.66663, 78.20052	30 KM/Hr
20	Turn	Medium	11.66698, 78.20136	30 KM/Hr
21	Turn	Medium	11.66886, 78.20279	30 KM/Hr
22	Turn	Medium	11.67280, 78.20401	30 KM/Hr
23	Turn	High	11.67332, 78.20333	15 KM/Hr
24	Turn	Medium	11.67389, 78.20336	30 KM/Hr
25	Turn	Medium	11.67583, 78.20611	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
23	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
25	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
26	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
27	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
28	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
29	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
30	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
31	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
32	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
33	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
37	hospital	Dr.Thiru Neuro Multi Speciality	11.6623376, 78.203353	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
24	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
34	school	St.joseph's matric hr.sec school	11.6419382, 78.1709055	30 km/h	Medium
35	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
36	school	Holy Cross Matriculation Higher Secondary School	11.6635657, 78.203532	30 km/h	Medium
38	school	Govt. High School, K.N.Colony	11.6617684, 78.1970397	30 km/h	Medium
39	school	Jothi Matriculation School	11.6622074, 78.1968379	30 km/h	Medium
40	school	Kavi Bharathi N&P School	11.6732663, 78.202141	30 km/h	Medium
41	school	Veeranam Primary School	11.6827435, 78.2155859	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45004, 77.87206



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61753, 78.10608



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64162, 78.12160



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64120, 78.12168



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65627, 78.20215



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.66372, 78.20649



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.66216, 78.19911



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66365, 78.19907



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66615, 78.20042



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66663, 78.20052



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66698, 78.20136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66886, 78.20279



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67280, 78.20401



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67332, 78.20333



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67389, 78.20336



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67583, 78.20611